



**FAR WESTERN UNIVERSITY
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SCHOOL OF ENGINEERING**



A Lab Report of Data Mining and Data Warehousing

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Lab 1

Title: Perform classification using Naive Bayes Algorithm and Bayes Net Algorithm and analyze.

Classification in Data Mining:

Classification is a **supervised learning** technique used to predict predefined **categorical class labels** for new data instances based on labeled training data. A classification model learns relationships between input attributes and class labels and applies this knowledge to unseen data.

Common Classification Algorithms

- Decision Trees
- Naive Bayes
- k-Nearest Neighbors (kNN)
- Support Vector Machines (SVM)
- Logistic Regression
- Random Forest
- Neural Networks

Naive Bayes and Bayes Net:

Naive Bayes is fast and simple, assuming independence among features. It works well for large datasets but may lose accuracy when features are correlated.

Bayes Net models dependencies between attributes using probabilistic graphs, providing better accuracy for complex data but requiring more computation.

Dataset Description:

The **Cards dataset** contains credit card and customer attributes such as card type, credit limit, number of cards owned, chip availability, and card-on-dark-web status. It is used to analyze customer behavior, credit risk, and fraud detection.

KDD Steps Followed

1. **Selection:** Relevant attributes like credit limit, card type, and security features were chosen.
2. **Pre-processing:** Missing values and duplicates were removed.
3. **Transformation:** Categorical attributes were encoded and data normalized.
4. **Data Mining:** Classification models such as Decision Trees and Naive Bayes were applied.
5. **Evaluation:** Models were evaluated using accuracy, precision, recall, F-measure, and MAE.
6. **Knowledge Representation:** Results were visualized using confusion matrices and decision trees.
7. **Decision-Making:** The insights support fraud detection and credit risk assessment.

Result:

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose NaiveBayes

Test options

☒ Use training set

☐ Supplied test set

☐ Cross-validation

☐ Percentage split

Set...

Folds10

%66

More options...

(Nom) card_brand

Start

Stop

Result list (right-click for options)

21:20:31 - bayes.NaiveBayes

21:21:00 - bayes.NaiveBayes

21:22:41 - bayes.NaiveBayes

Classifier output

Time taken to build model: 0.03 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0.05 seconds

=== Summary ===

Correctly Classified Instances

6096

99.1865 %

Incorrectly Classified Instances

50

0.8135 %

Kappa statistic

0.9859

Mean absolute error

0.0195

Root mean squared error

0.0719

Relative absolute error

6.7415 %

Root relative squared error

18.8927 %

Total Number of Instances

6146

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.993	0.006	0.991	0.993	0.992	0.987	1.000	1.000	Visa
	0.993	0.010	0.991	0.993	0.992	0.984	1.000	1.000	Mastercard
	0.947	0.000	1.000	0.947	0.973	0.972	1.000	0.999	Discover
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Amex
Weighted Avg.	0.992	0.007	0.992	0.992	0.992	0.985	1.000	1.000	

=== Confusion Matrix ===

a	b	c	d	<-- classified as
2309	17	0	0	a = Visa
22	3187	0	0	b = Mastercard
0	11	198	0	c = Discover
0	0	0	402	d = Amex

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose BayesNet -D -Q weka.classifiers.bayes.net.search.local.K2 -- -P 1 -S BAYES -E weka.classifiers.bayes.net.estimate.SimpleEstimator -- -A 0.5

Test options

☒ Use training set

☐ Supplied test set

☐ Cross-validation

☐ Percentage split

Set...

Folds10

%66

More options...

(Nom) card_brand

Start

Stop

Result list (right-click for options)

21:20:31 - bayes.NaiveBayes

21:21:00 - bayes.NaiveBayes

21:22:41 - bayes.NaiveBayes

21:23:13 - bayes.BayesNet

Classifier output

Time taken to build model: 0.22 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0.1 seconds

=== Summary ===

Correctly Classified Instances

6146

100 %

Incorrectly Classified Instances

0

0 %

Kappa statistic

1

Mean absolute error

0.0002

Root mean squared error

0.0008

Relative absolute error

0.0716 %

Root relative squared error

0.2074 %

Total Number of Instances

6146

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Visa
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Mastercard
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Discover
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Amex
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	

=== Confusion Matrix ===

a	b	c	d	<-- classified as
2326	0	0	0	a = Visa
0	3209	0	0	b = Mastercard
0	0	209	0	c = Discover
0	0	0	402	d = Amex

Comparison:

Naive Bayes provided fast classification with moderate accuracy, while Bayes Net achieved better accuracy by modeling feature dependencies. Decision Trees offered clear interpretability of classification rules.

Conclusion:

Classification techniques effectively predict credit card brand and usage patterns. Naive Bayes is suitable for quick and simple analysis, while Bayes Net is preferred when attribute relationships are important. Overall performance can be improved using feature selection and advanced models.

Lab 2:

Title: Perform Clustering by using simple K-means Algorithm and Hierarchical algorithm and analyze:

Clustering In Data Mining

Clustering is an unsupervised data mining technique used to group similar data objects without predefined class labels. It helps in discovering hidden patterns and relationships in data and is widely used in customer segmentation, fraud detection, and risk analysis.

Simple k-Means and Density Based Clustering:

kMeans Clustering is a partition-based clustering algorithm that divides data into k clusters by assigning points to the nearest centroid and updating centroids iteratively. It is fast, efficient, and works well for large datasets with well-separated clusters. However, it requires the number of clusters to be specified in advance and is sensitive to noise and outliers.

Density-Based Clustering (DBSCAN) groups data points based on dense regions using parameters Eps and MinPts. It does not require the number of clusters beforehand and can identify arbitrarily shaped clusters and noise. DBSCAN is effective for anomaly detection but may struggle with datasets with varying densities.

Result :

The screenshot displays the Weka software interface, specifically the 'Clusterer' tab. The 'SimpleKMeans' algorithm is selected. The 'Cluster mode' section shows 'Use training set' is selected. The 'Ignore attributes' section is empty. The 'Start' button is visible. The 'Result list (right-click for options)' shows '02.02.20 - SimpleKMeans' selected. The 'Clusterer output' pane shows the following results:

Clusterer output

SimpleKMeans

Number of iterations: 21
Within cluster sum of squared errors: 25306.848409267986

Initial starting points (random):

Cluster 0: 670,1633,Mastercard,Debit,5773754472145445,08/2024,145,YES,2,521982,01/2020,2020,Mo
Cluster 1: 6101,421,Mastercard,Debit,5865665247394120,08/2024,626,YES,2,216094,12/2014,2014,Mo

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data (6146.0)	Cluster# 0 (1601.0)	1 (4545.0)
id	3072.5	915.604	3832.2779
client_id	994.9396	978.8401	1000.6108
card_brand	Mastercard	Mastercard	Mastercard
card_type	Debit	Debit	Debit
card_number	4820425803048972	4795561403007901	4827775555553428
expires	02/2020	02/2020	02/2020
crr	506.2208	523.1693	500.2506
has_chip	YES	YES	YES
num_cards_issued	1.5031	1.5009	1.5039
credit_limit	\$0	\$8700	\$0
acct_open_date	02/2020	01/2020	02/2020
year_pin_last_changed	2013.4367	2015.2711	2012.7905
card_on_dash_web	No	No	No

Time taken to build model (full training data) : 0.22 seconds

=== Model and evaluation on training set ===

Clustered Instances

Cluster	Count	Percentage
0	1601	(26%)
1	4545	(74%)

Preprocess Classify **Cluster** Associate Select attributes Visualize

Cluster

Choose **MakeDensityBasedClusterer** -M 1.0E-6 -W weka.clusters.SimpleKMeans --init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (nom) card_on_dark_web
☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

00.1002 - MakeDensityBasedClusterer

Cluster output

Number of iterations: 21
Within cluster sum of squared errors: 25306.848409267986
Initial starting points (random):
Cluster 0: 670,1639,Mastercard,Debit,5773754472145445,06/2024,145,YES,2,021982,01/2020,2020,No
Cluster 1: 6101,421,Mastercard,Debit,5855655247394120,09/2024,626,YES,2,016094,12/2014,2014,No
Missing values globally replaced with mean/mode
Final cluster centroids:

Attribute	Full Data (6146.0)	Cluster# 0 (1601.0)	Cluster# 1 (4545.0)
id	3072.5	915.604	3832.2779
client_id	994.9396	978.8401	1000.6108
card_brand	Mastercard	Mastercard	Mastercard
card_type	Debit	Debit	Debit
card_number	4820425803848972	4799561409307901	4827775359553428
expires	02/2020	02/2020	02/2020
cvv	506.2208	523.1693	500.2506
has_chip	YES	YES	YES
num_cards_issued	1.5031	1.5009	1.5039
credit_limit	\$0	\$8700	\$0
acct_open_date	02/2020	01/2020	02/2020
year_pin_last_changed	2013.4367	2015.2711	2012.7905
card_on_dark_web	No	No	No

Fitted estimators (with ML estimates of variance):
Cluster: 0 Prior probability: 0.2606
Attribute: id
Normal Distribution. Mean = 915.604 StdDev = 698.328

Preprocess Classify **Cluster** Associate Select attributes Visualize

Cluster

Choose **MakeDensityBasedClusterer** -M 1.0E-6 -W weka.clusters.SimpleKMeans --init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (nom) card_on_dark_web
☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

00.1002 - MakeDensityBasedClusterer

Cluster output

Attribute: id
Normal Distribution. Mean = 915.604 StdDev = 698.328
Attribute: client_id
Normal Distribution. Mean = 978.8401 StdDev = 571.6177
Attribute: card_brand
Discrete Estimator. Counts = 619 826 54 106 (Total = 1605)
Attribute: card_type
Discrete Estimator. Counts = 891 551 162 (Total = 1604)
Attribute: card_number
Normal Distribution. Mean = 4799561409307901 StdDev = 1328326944458587.0
Attribute: expires
Discrete Estimator. Counts = 26 22 19 16 1 2 29 25 29 22 18 24 114 27 14 3 25 29 4 19 14 19 18 21 23 4 22 17 25 22 3 14 27 25 4 12 26 14 2 3 22 7 27 21 20 2 83
Attribute: cvv
Normal Distribution. Mean = 523.1693 StdDev = 291.8771
Attribute: has_chip
Discrete Estimator. Counts = 1425 178 (Total = 1603)
Attribute: num_cards_issued
Normal Distribution. Mean = 1.5009 StdDev = 0.5244
Attribute: credit_limit
Discrete Estimator. Counts = 1 1 1 4 2 1 2 1 7 1 1 1 2 1 1 2 4 1 1 5 1 3 1 6 1 2 1 1 3 26 1 1 1 1 2 1 2 1 5 1 3 5 4 1 4 2 2 1 1 3 1 2 1 2 2 2 1 2 4 1 2 1 9 1
Attribute: acct_open_date
Discrete Estimator. Counts = 1 1 1 15 2 1 32 1 1 7 14 1 29 34 1 1 9 1 1 1 2 572 2 1 25 17 1 1 6 28 1 1 1 1 15 41 1 1 1 20 1 34 12 4 0 1 1 1 3 1 1 1 1 1 1
Attribute: year_pin_last_changed
Normal Distribution. Mean = 2015.2711 StdDev = 4.4715
Attribute: card_on_dark_web
Discrete Estimator. Counts = 1602 (Total = 1602)
Cluster: 1 Prior probability: 0.7394
Attribute: id
Normal Distribution. Mean = 3832.2779 StdDev = 1367.0373
Attribute: client_id
Normal Distribution. Mean = 1000.6108 StdDev = 580.8896
Attribute: card_brand
Discrete Estimator. Counts = 1709 2385 157 298 (Total = 4549)
Attribute: card_type
Discrete Estimator. Counts = 2622 1508 418 (Total = 4548)
Attribute: card_number
Normal Distribution. Mean = 4827775359553428 StdDev = 1328447910234379
Attribute: expires
Discrete Estimator. Counts = 67 53 67 57 3 2 53 60 35 63 60 265 61 60 7 58 65 7 52 55 43 67 74 59 13 67 65 50 61 4 58 68 57 6 59 53 51 10 5 73 9 56 61 61 2 4
Attribute: cvv
Normal Distribution. Mean = 500.2506 StdDev = 288.2954
Attribute: has_chip
Discrete Estimator. Counts = 4077 470 (Total = 4547)
Attribute: num_cards_issued
Normal Distribution. Mean = 1.5039 StdDev = 0.5173
Attribute: credit_limit
Discrete Estimator. Counts = 2 2 2 8 5 3 1 2 4 3 2 2 2 2 1 12 2 2 2 15 2 7 2 16 2 1 2 2 6 1 2 2 2 2 1 2 1 2 12 2 11 4 12 2 11 1 1 2 2 5 2 18 2 1 1 1 2 2 6 2
Attribute: acct_open_date
Discrete Estimator. Counts = 13 19 22 1 50 18 1 47 47 42 7 1 52 2 1 49 8 1 25 27 25 51 17 1 37 15 1 1 10 46 603 1 35 15 42 8 16 1 1 20 14 9 1 39 2 1 1 25 16 11
Attribute: year_pin_last_changed
Normal Distribution. Mean = 2012.7905 StdDev = 4.0017
Attribute: card_on_dark_web
Discrete Estimator. Counts = 4546 (Total = 4546)
Time taken to build model (full training data) : 0.3 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 1508 (25%)
1 4638 (75%)
Log likelihood: -82.73806

Comparison:

When applied to the Cards dataset in WEKA, K-Means produced compact and well-defined clusters . DBSCAN identified irregular clusters and detected noisy or risky customer profiles that K-Means could not isolate.

Conclusion:

K-Means is suitable for large datasets with clear patterns due to its efficiency and simplicity. Density-Based Clustering is better for detecting anomalies and complex cluster structures. The choice of algorithm depends on dataset size, structure, and analysis objectives.